EXERCISE 1 [7]

a. Expand $\left(\frac{4}{5}+\frac{1}{5}\right)^5$. [2]

b. With every attempt, Jack has an 80% chance of kicking a goal. In one quarter of a match he has 5 kicks for goal.

Determine the probability that he scores :

i. 3 goals then misses twice [2]

ii. 3 goals and misses twice. [3]

EXERCISE 2 [6]

In a game, the numbers from 1 to 20 are written on tickets and placed in a bag. A player draws out a number at random. He or she wins \$3 if the number is even, \$6 if the number is a square number, and \$9 if the number is both even and square.

a. Calculate the probability that the player wins : [3]

i. \$3

ii. \$6

iii. \$9

b. How much should be charged to play the game so that it is a fair game ? [3]

Answer Key

Exercise 1

a. $\left(\frac{4}{5}+\frac{1}{5}\right)^5 = \left(\frac{4}{5}\right)^5 + 5\left(\frac{4}{5}\right)^4\left(\frac{1}{5}\right) + 10\left(\frac{4}{5}\right)^3\left(\frac{1}{5}\right)^2 + 10\left(\frac{4}{5}\right)^2\left(\frac{1}{5}\right)^3 + 5\left(\frac{4}{5}\right)\left(\frac{1}{5}\right)^4 + \left(\frac{1}{5}\right)^5$

b.i. $\mathrm{P}(\text{GGGMM}) = \left(\frac{4}{5}\right)^3 \times \left(\frac{1}{5}\right)^2 = \frac{64}{3125}$

b.ii. $\mathrm{P}(X=3) = \binom{5}{3}\left(\frac{4}{5}\right)^3\left(\frac{1}{5}\right)^2 = 10 \times \frac{64}{3125} = \frac{640}{3125} \approx 0.205$

Exercise 2

a.i. Even numbers (excluding square-and-even) : $\{2,6,8,10,12,14,18,20\}$, so $\mathrm{P}(\$3) = \frac{8}{20} = \frac{2}{5}$

a.ii. Square numbers (excluding even squares) : $\{1,9,25\} \cap [1,20] = \{1,9\}$, so $\mathrm{P}(\$6) = \frac{2}{20} = \frac{1}{10}$

a.iii. Even and square : $\{4,16\}$, so $\mathrm{P}(\$9) = \frac{2}{20} = \frac{1}{10}$

b. $\mathrm{E}(\text{winnings}) = 3 \times \frac{2}{5} + 6 \times \frac{1}{10} + 9 \times \frac{1}{10} = \frac{6}{5} + \frac{6}{10} + \frac{9}{10} = \frac{12}{10} + \frac{15}{10} = \frac{27}{10} = \$2.70$

The game is fair if the entry fee is \$2.70.